

Renal Replacement Therapies

SL Lui

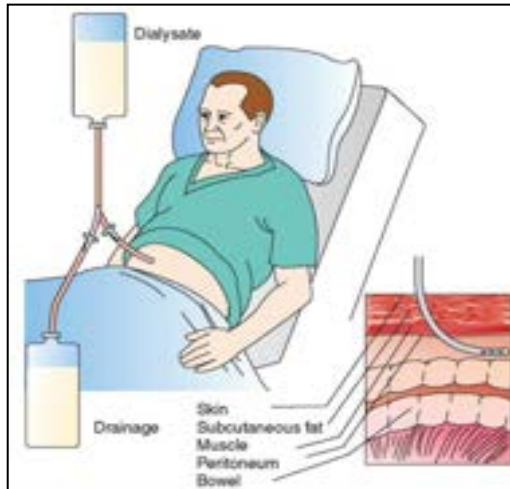
Department of Medicine

TWH & QMH

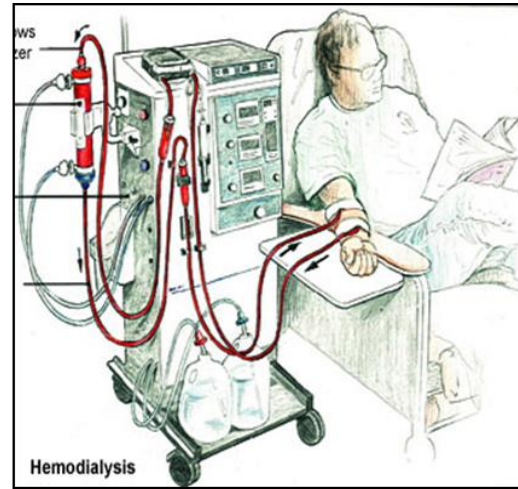
Renal Replacement Therapy

- **Renal replacement therapy (RRT) refers to therapy that replaces the functions of the failed kidneys in patients with end stage kidney disease (ESKD)**

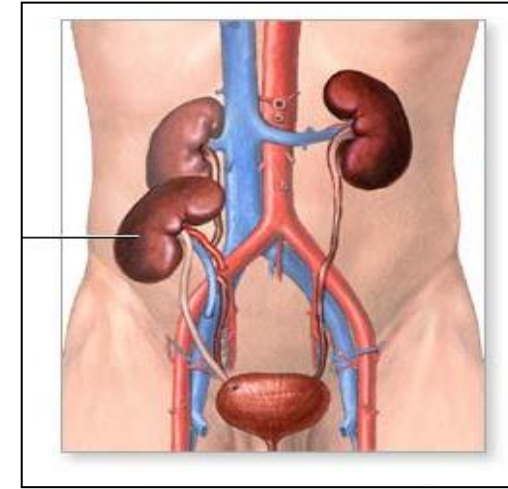
Three main forms of RRT



**Peritoneal Dialysis
(PD)**



**Hemodialysis
(HD)**

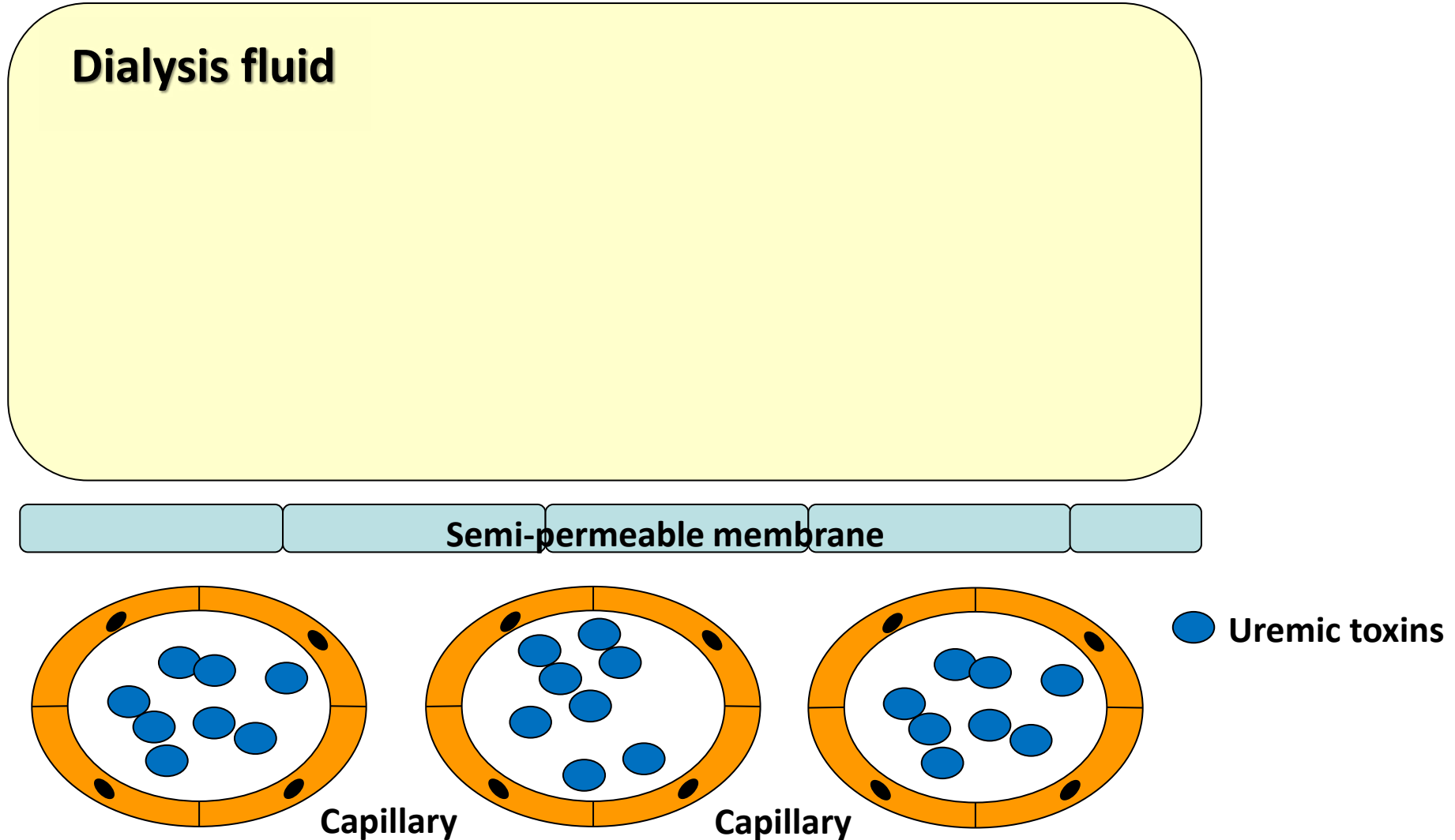


**Kidney
Transplantation**

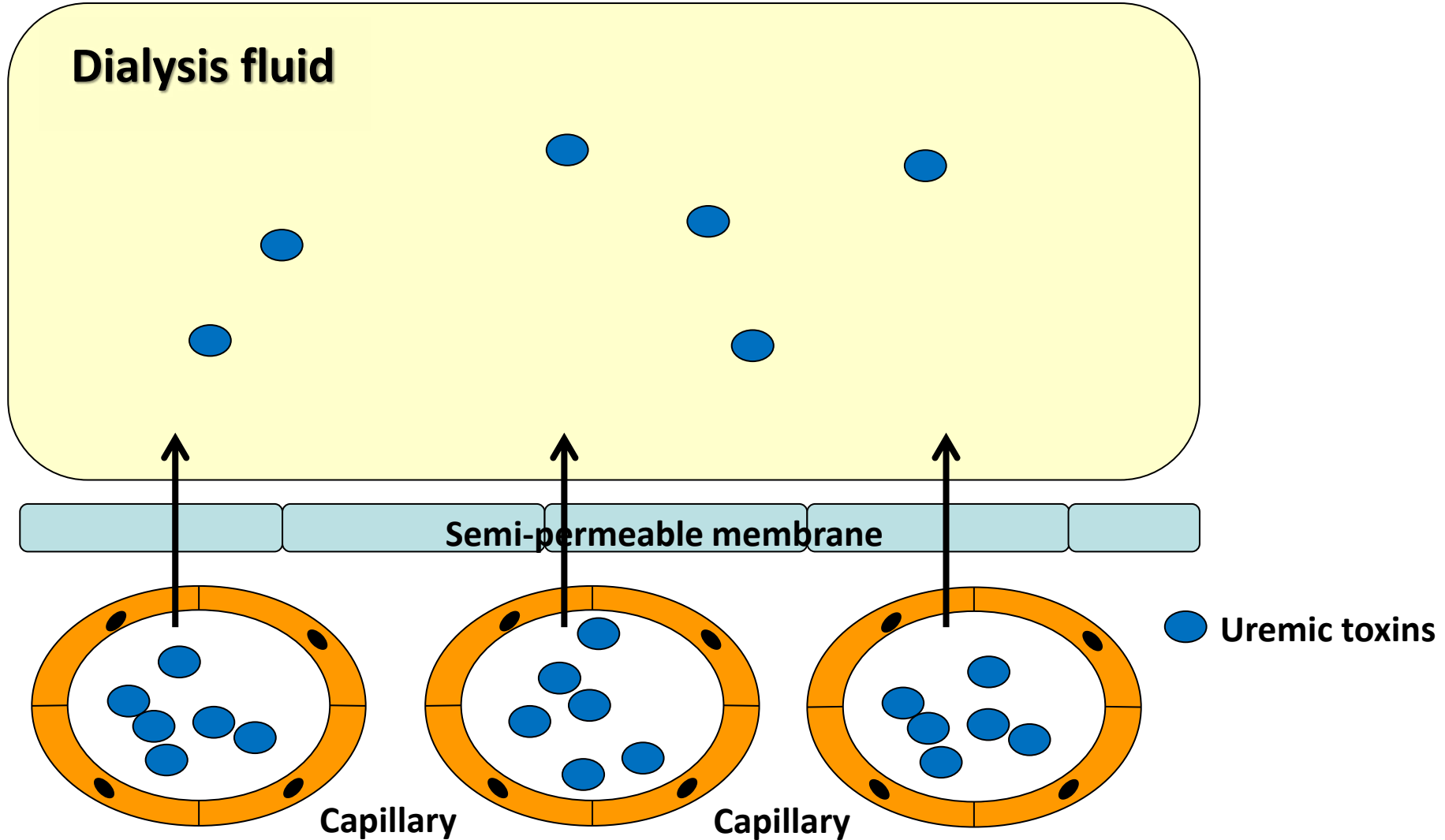
Initiation of RRT

- Initiation of RRT should be considered in ESKD patients when
 1. their glomerular filtration rate (GFR) falls to <10 ml/min/1.73m²
 2. they develop uremic symptoms

Basic principles of dialysis



Basic principles of dialysis

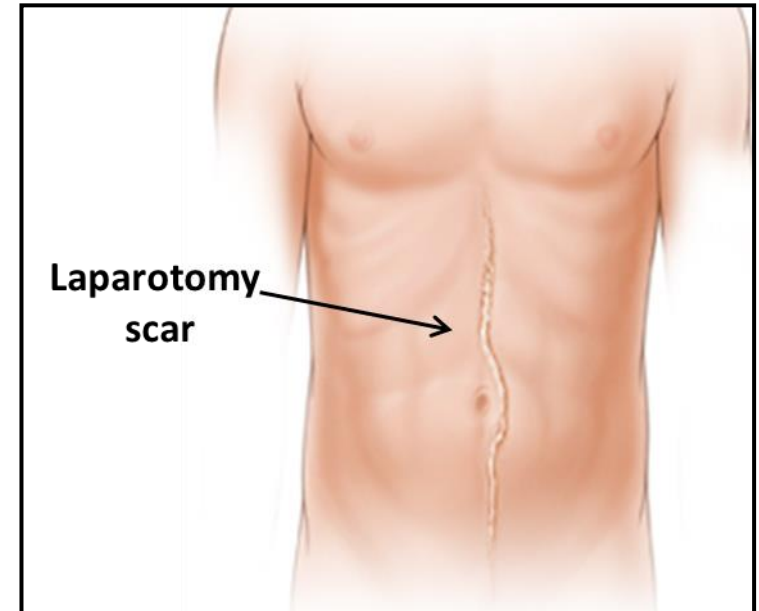


Choice between PD & HD

- **Factors to be considered when choosing between PD and HD for patients with ESRD include:**
 - **whether the patient has any contraindications to either PD or HD**
 - **local health care reimbursement system**
 - **patient's preference**

Contraindications to PD

- **History of major abdominal surgery which caused extensive scarring of the peritoneal membrane.**



Contraindications to HD

- **Poor cardiac function such that the patient is unable to tolerate the hemodynamic changes during HD**
- **Lack of suitable vascular access for HD**

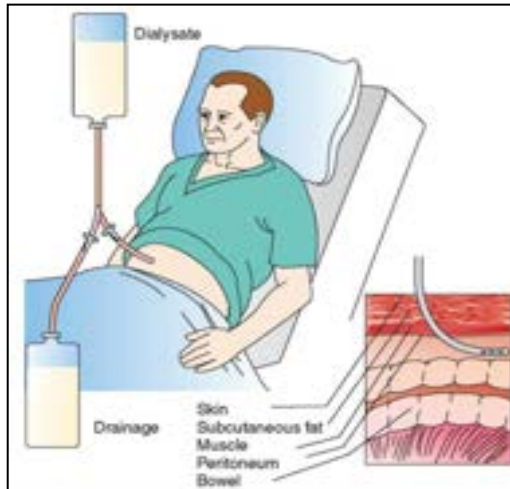
Local health care reimbursement system

- **The local healthcare reimbursement system often plays an determining role in the choice between PD & HD**
- **Hong Kong's public health care system has adopted a PD first policy because PD is less costly**
- **ESKD patients will be offered PD first in HA hospitals unless they have contraindications to PD**

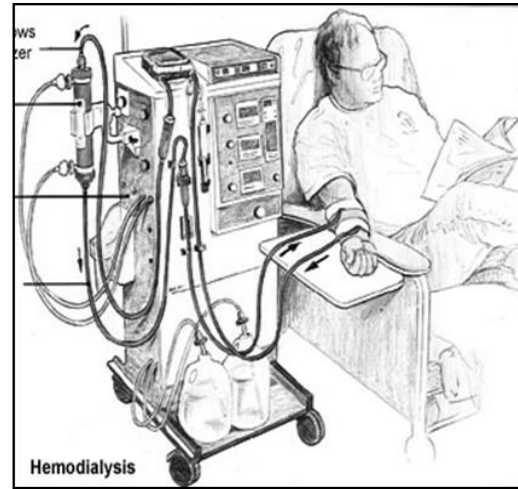
Patient's preference

- Peritoneal dialysis is usually performed by the patient at home on a daily basis
- Hemodialysis is usually performed by the renal nurses in a HD center 2-3 times per week
- ESKD patients may prefer one form of dialysis over the other due to personal or social reasons

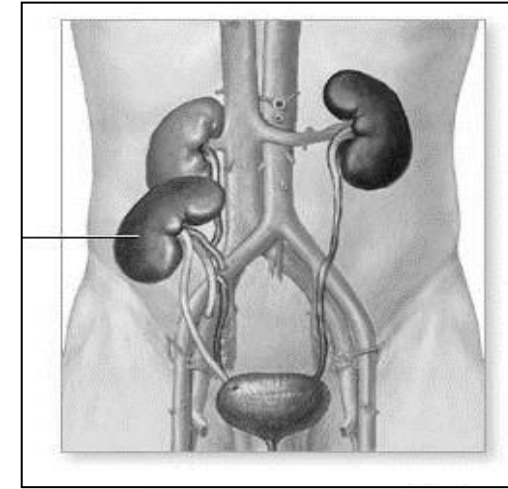
Three main forms of RRT



**Peritoneal Dialysis
(PD)**



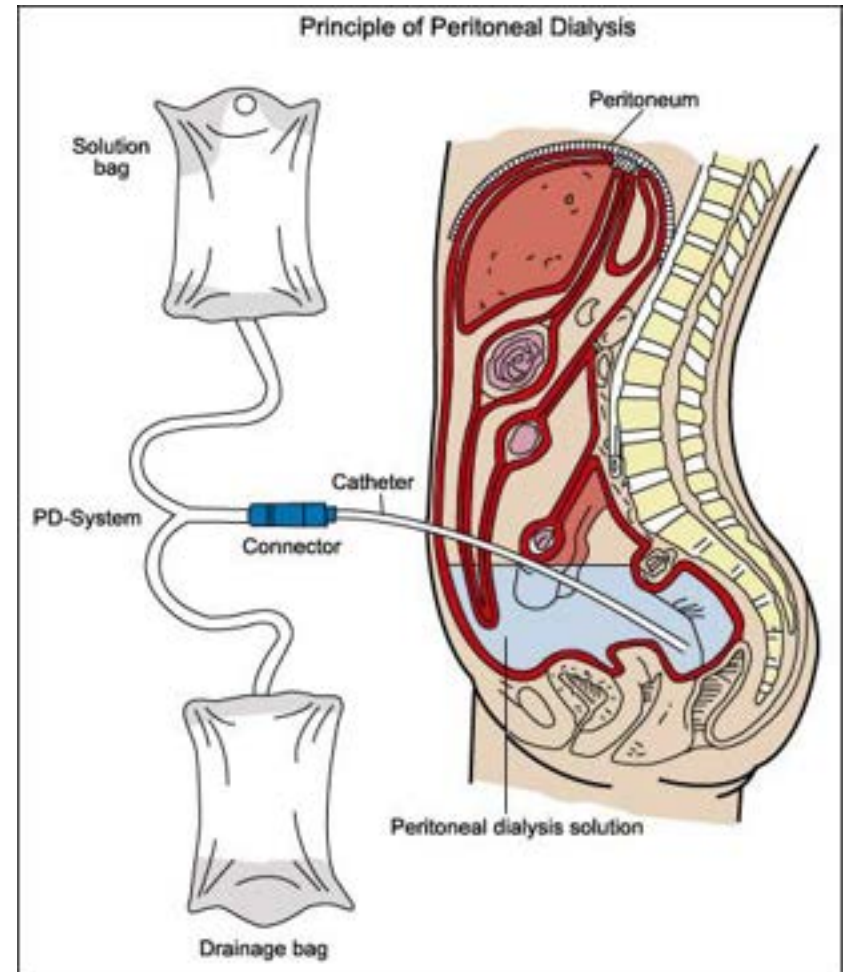
**Hemodialysis
(HD)**



**Kidney
Transplantation**

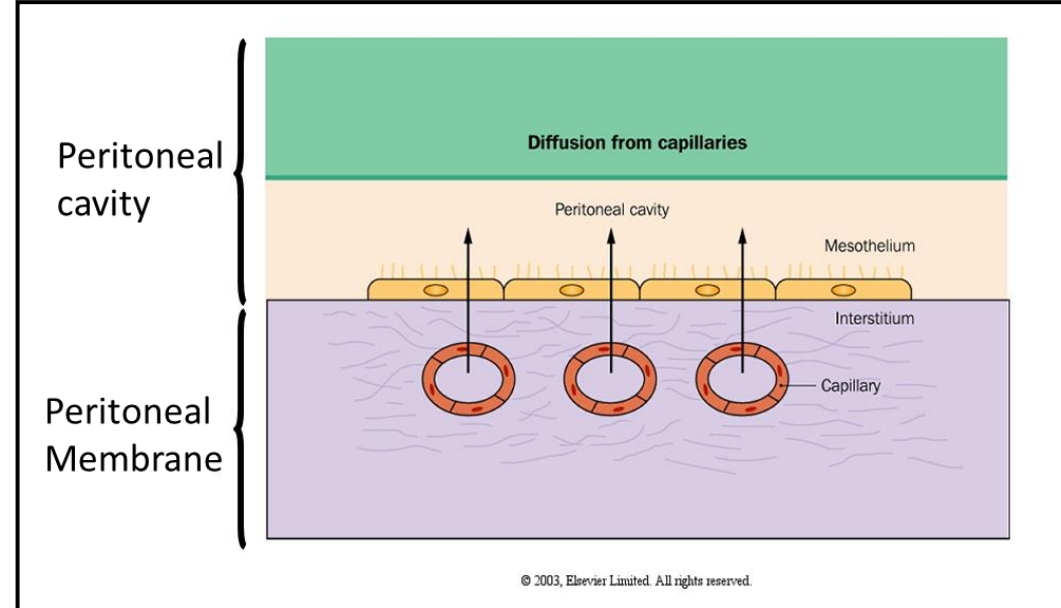
Essential components of PD

- **Peritoneal membrane**
- **Peritoneal dialysis catheter**
- **Peritoneal dialysis fluids**



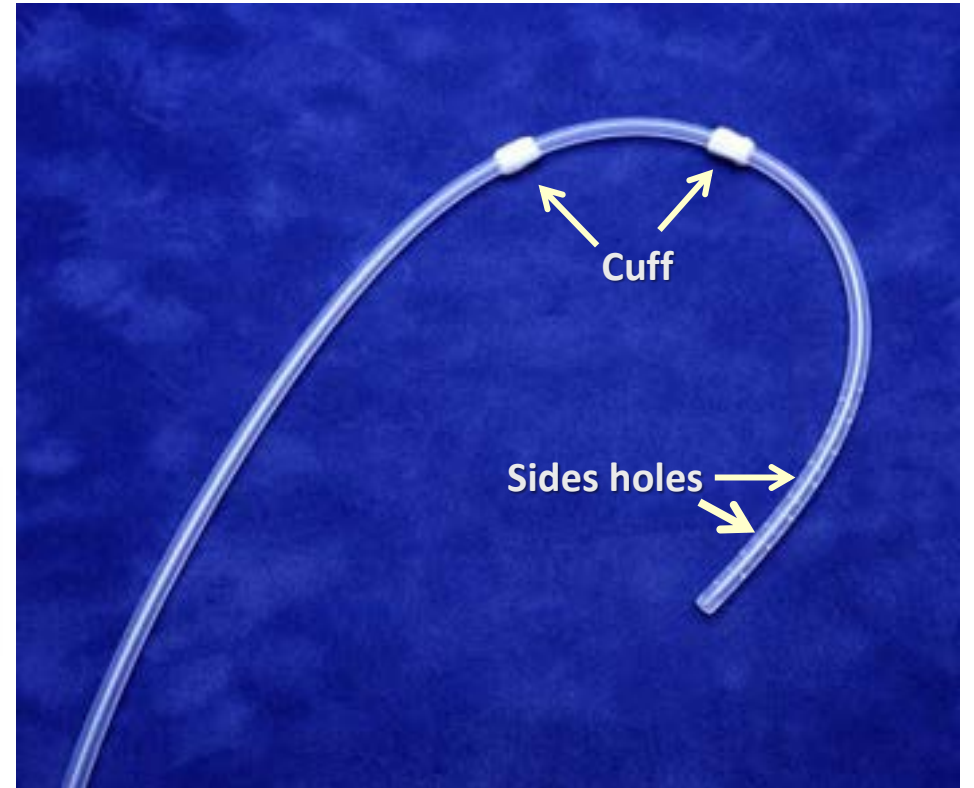
Peritoneal membrane

- PD makes use of the peritoneal membrane as the dialysis membrane because it is thin, vascularized and has a large surface area

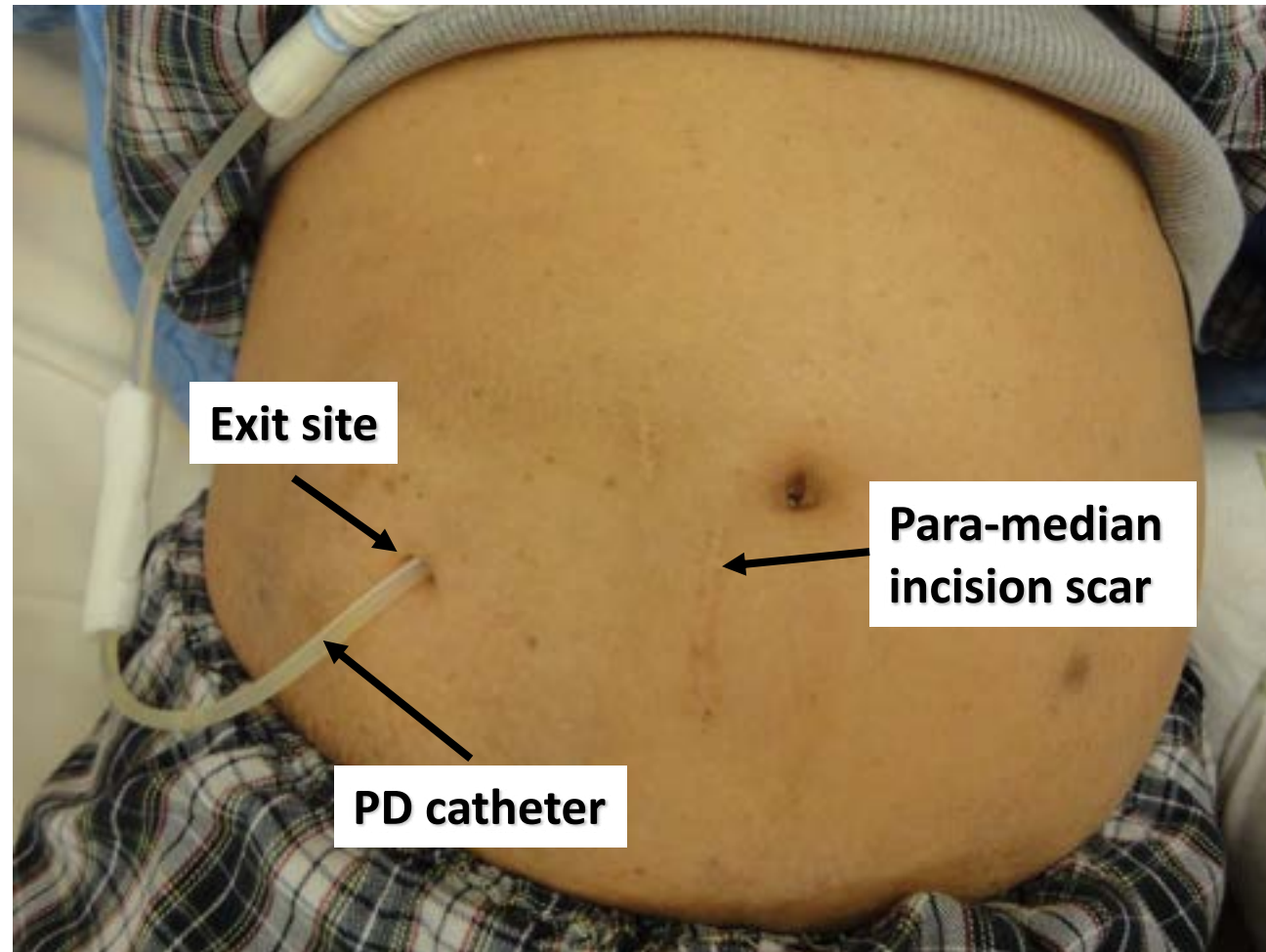


Peritoneal dialysis catheter

- The peritoneal dialysis catheter (commonly known as the Tenckhoff catheter) is a specially designed plastic tube with many side-holes in one end and two fibrous cuffs in the other end



Peritoneal dialysis catheter



Peritoneal dialysis catheter

- Ideally, the tip of the PD catheter should be located inside the pelvis pointing downwards

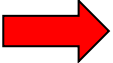
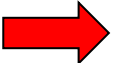


Peritoneal Dialysis Fluids (PDF)

- Commonly used PDFs are balanced salt solutions, with glucose as the osmotic agent and lactate as the buffer
- They do not contain potassium

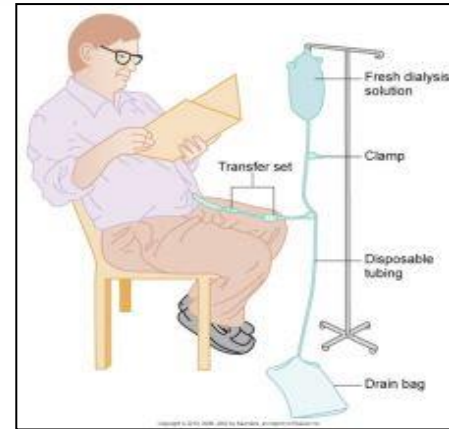


Typical composition of PDF

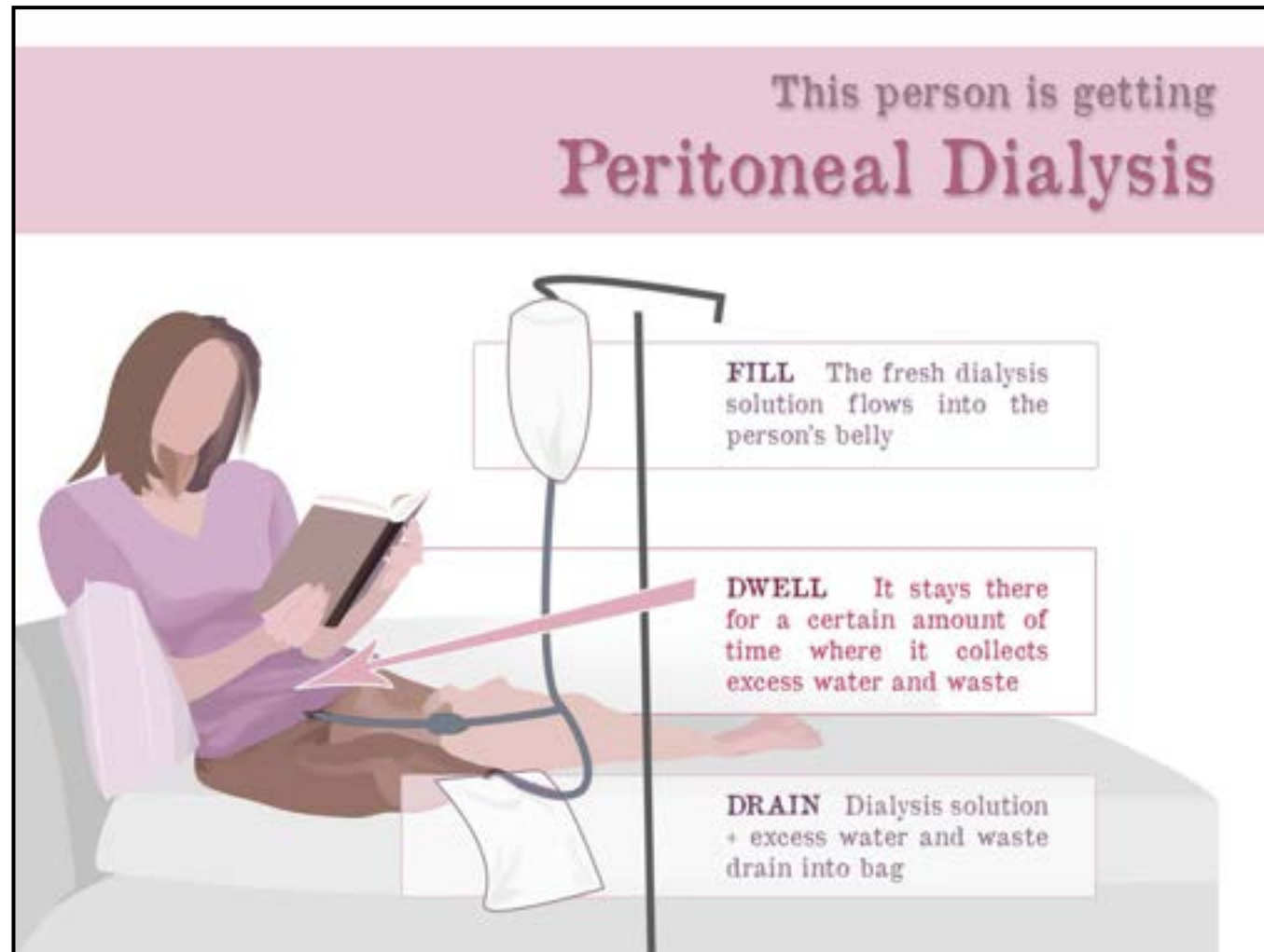
	Volume (Liter)	2
	Glucose (g/dL)	1.5 / 2.5 / 4.25
	Potassium (mmol/L)	0
	Sodium (mmol/L)	132
	Calcium (mmol/L)	2.5 - 3.5
	Magnesium (mmol/L)	0.5 - 1.5
	Lactate (mmol/L)	35 – 40

Two main forms of PD

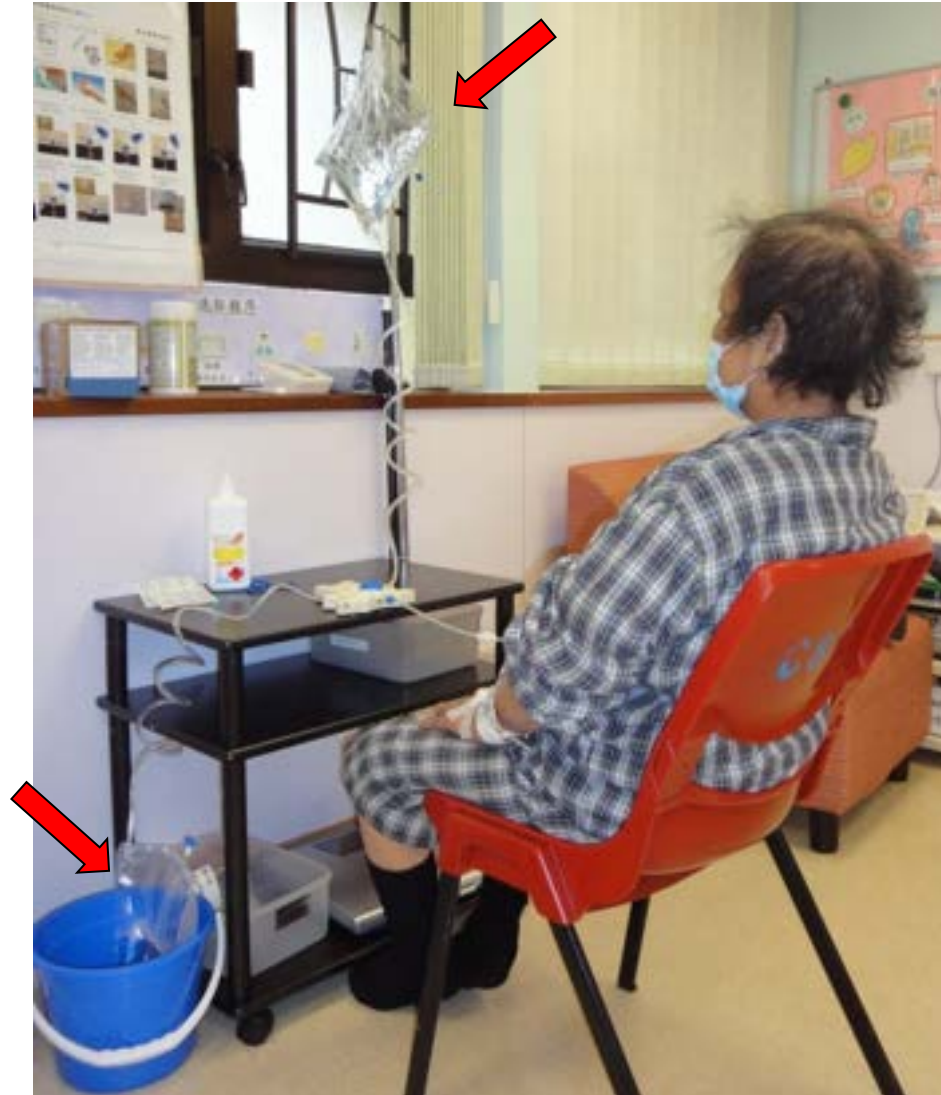
- **Continuous ambulatory peritoneal dialysis (CAPD)**
- **Automated peritoneal dialysis (APD)**



Continuous ambulatory peritoneal dialysis



Continuous ambulatory peritoneal dialysis



Automated peritoneal dialysis (APD)

- **Automated peritoneal dialysis makes use of an APD machine to help the patient to perform the PD exchanges, usually at night when the patient goes to bed**

Automated peritoneal dialysis



APD machine



Comparison between CAPD & APD

	CAPD	APD
Need an APD machine	No	Yes
Timing of PD exchange	Throughout the day	At night during sleep
Need for daytime exchange	Yes	No for NIPD ¹ Yes for CCPD ²
➡ Dialysis efficacy	Similar	
➡ Quality of life	Usually better in APD	
➡ Cost	\$\$	\$\$\$

¹NIPD: Nocturnal intermittent PD; ²CCPD: Continuous Cyclic PD

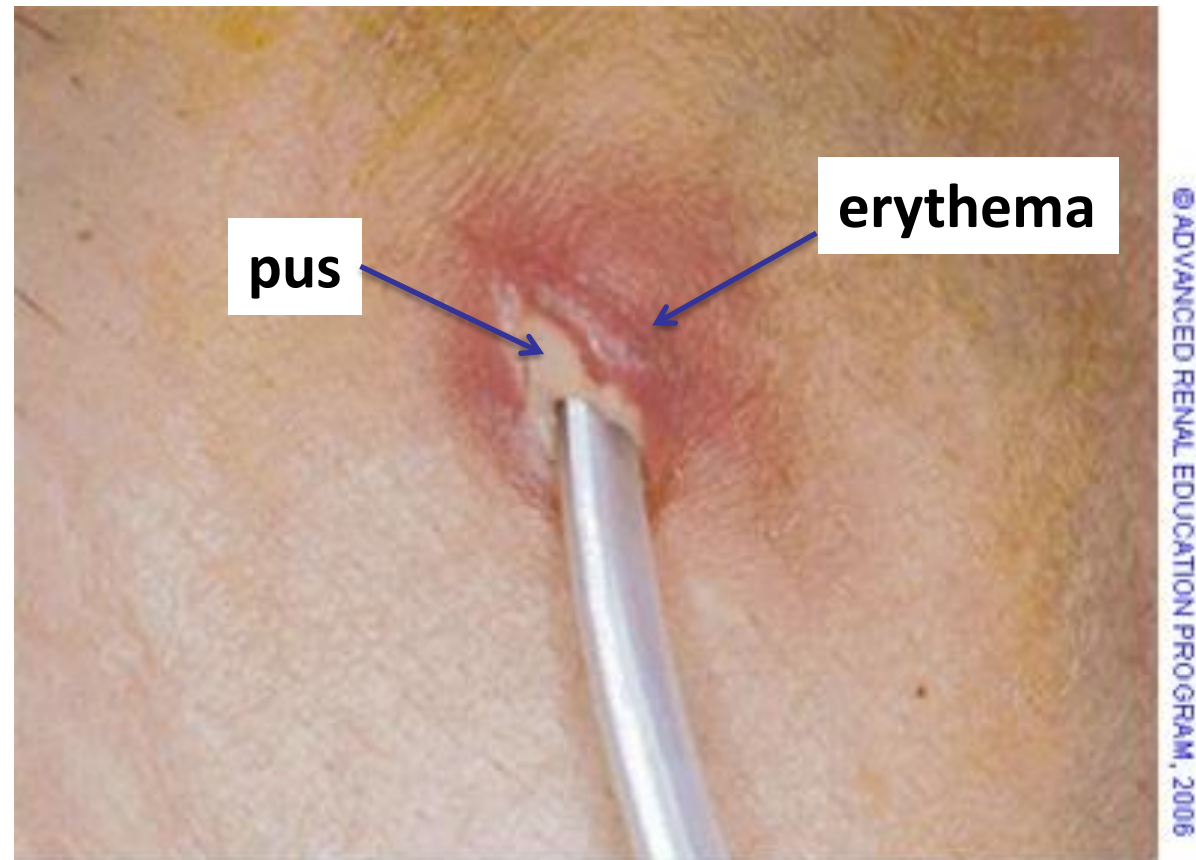
Complications of PD

- 1. PD catheter exit site infection**
- 2. PD related peritonitis**
- 3. Peritoneal dialysis failure**
- 4. PD catheter mal-function**
- 5. Peritoneal leakage**

PD catheter exit site infection

- Usually presents as redness and tenderness at the PD catheter exit site, sometimes associated with purulent discharge
- PD catheter exit site infection may predispose to PD peritonitis

PD catheter exit site infection



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Photo courtesy of ZJ Twardowski

PD related peritonitis

- The main presenting symptoms of PD related peritonitis are abdominal pain, turbid PD fluid and fever



PD related Peritonitis

- **Causative organisms:**
 - usually gram positive or negative bacteria
 - occasionally fungi or mycobacteria
- **Predisposing factors:**
 - exit site infection
 - contamination of the PD catheter during PD exchanges
 - constipation or diarrhea

PD related Peritonitis

- **Diagnosis:**
 - PD fluid microscopy shows elevated WBC count with neutrophil predominance
 - PD fluid culture identifies the causative organism
- **Treatment:**
 - empirical intraperitoneal antibiotics that cover common gram +ve and –ve bacteria (e.g. Cefazolin + Amikacin)

PD related Peritonitis

- **Treatment (continue):**
 - **Choice of antibiotics should be adjusted according to the PD fluid culture results**
 - **Patients with refractory peritonitis might need to have their PD catheter removed and be put on temporary hemodialysis**

PD related Peritonitis

- **Complications:**
 - Frequent PD related peritonitis will damage the peritoneal membrane & eventually lead to peritoneal dialysis failure

Peritoneal dialysis failure

- Chronic exposure to dextrose containing PD fluids and repeated peritonitis will cause fibrosis of the peritoneal membrane, eventually leading to PD failure



Normal peritoneum



After 10 years on PD, peritoneum became markedly thickened

PD catheter malposition

- Ideally, the tip of the PD catheter should be inside the pelvic cavity
- Occasionally, the tip of the PD catheter may migrate out of the pelvis, leading to poor drainage of the PD fluid

PD catheter malposition

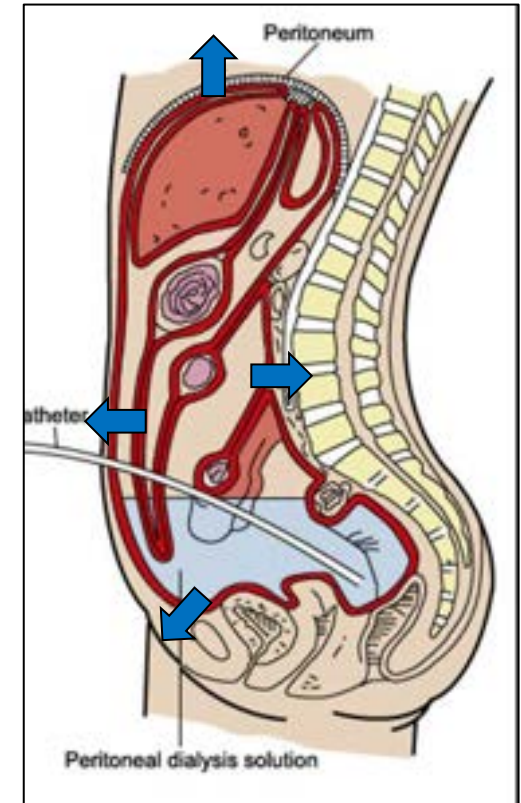


Peritoneal leakage

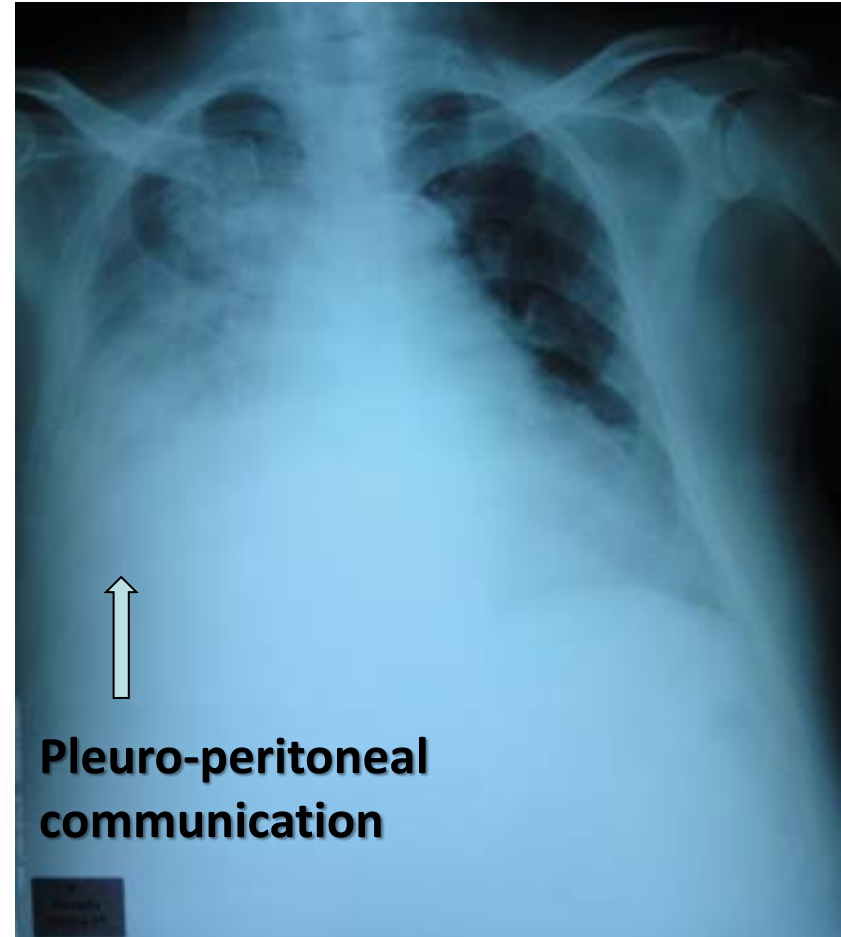
- **Normally, the PD fluid is contained inside the peritoneal cavity**
- **Occasionally, the PD fluid may leak out of the peritoneal cavity into the pleural cavity, the anterior abdominal wall or the retro-peritoneal space**

Peritoneal leakage

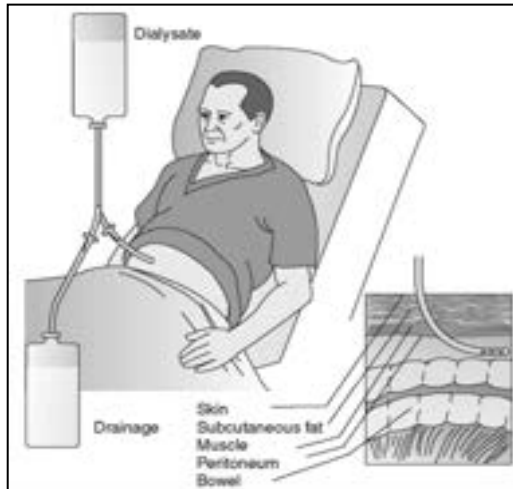
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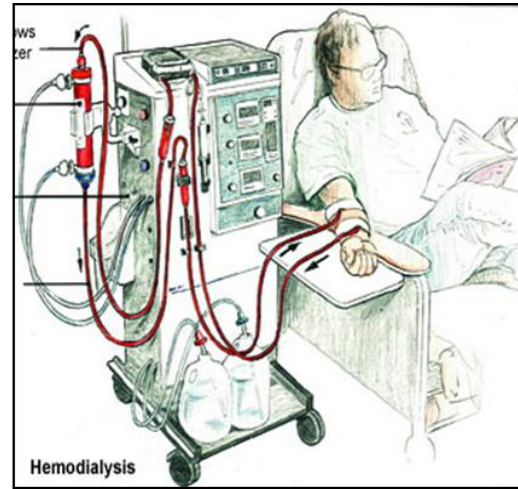
Peritoneal leakage



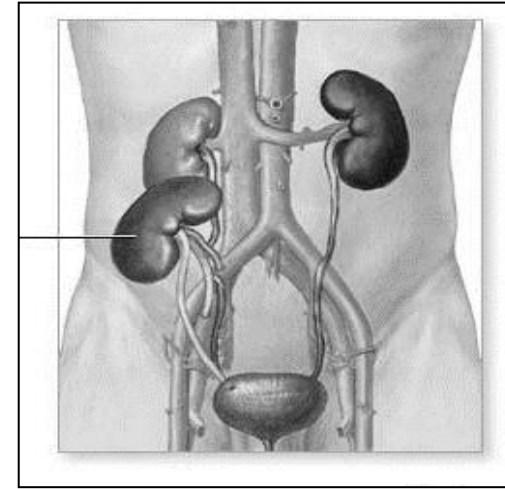
Three main forms of RRT



**Peritoneal Dialysis
(PD)**



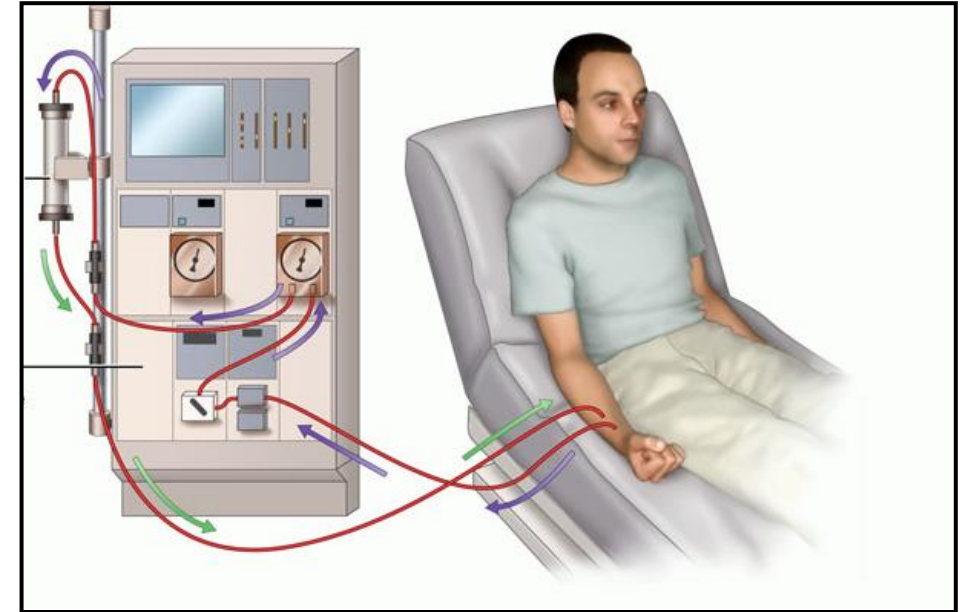
**Hemodialysis
(HD)**



**Kidney
Transplantation**

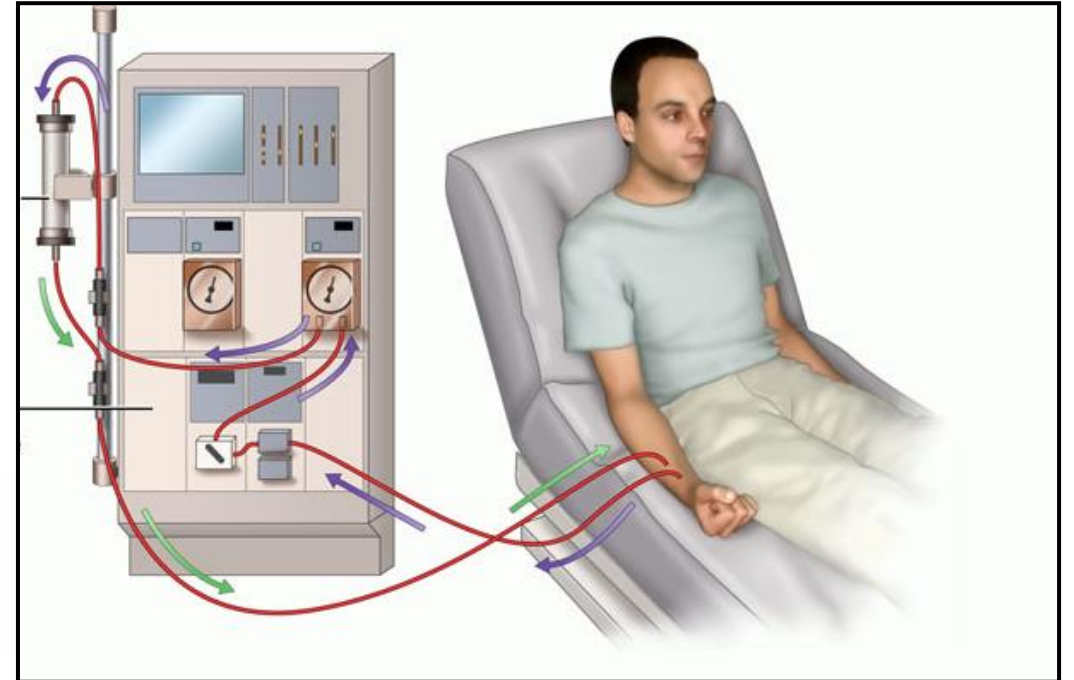
Essential components of Hemodialysis

- **Vascular access**
- **Hemodialysis machine**
- **Hemodialyzer**



Vascular access for HD

- In hemodialysis, a vascular access is required to allow sufficiently high flow of blood from the patient to pass through the hemodialysis machine and then return to the patient



Three main types of vascular access

1. **Arteriovenous fistula (AVF)**
2. **Arteriovenous graft (AVG)**
3. **Tunneled cuffed double lumen central venous catheter**

AV fistula

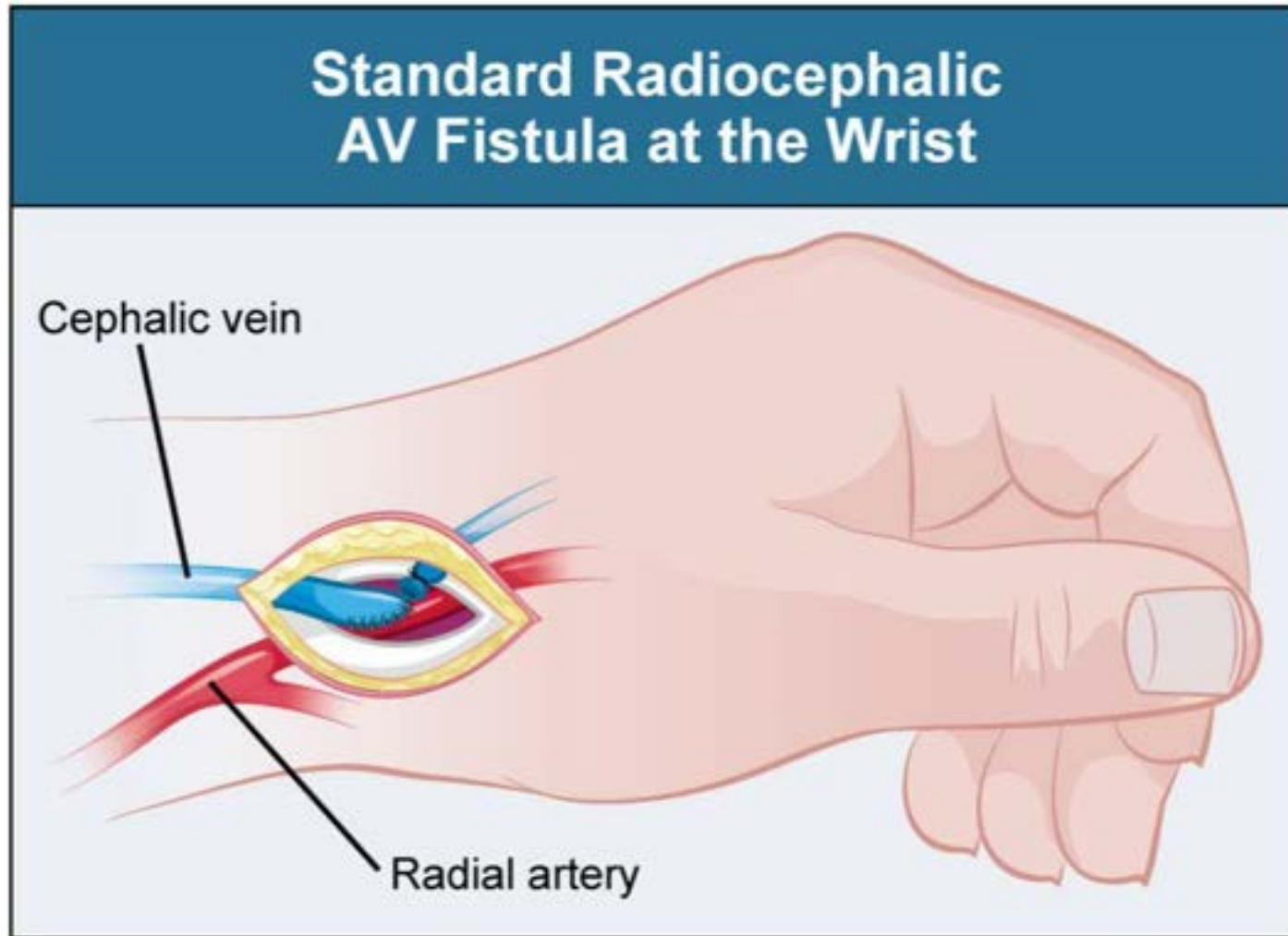
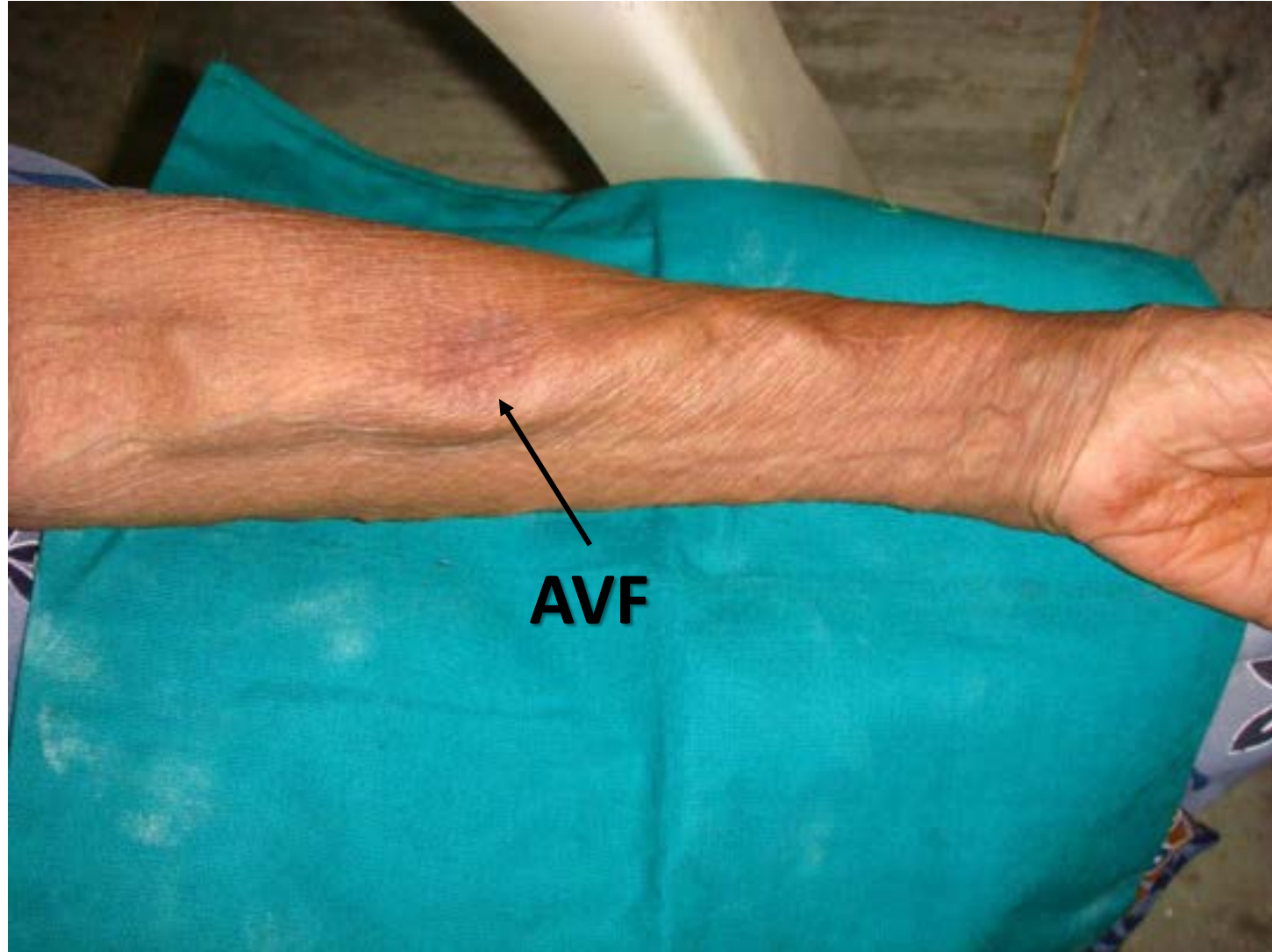
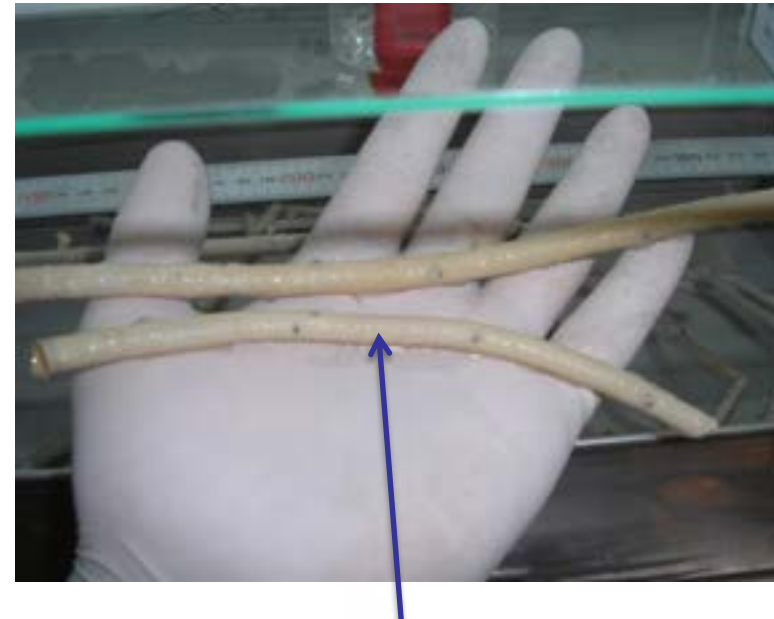
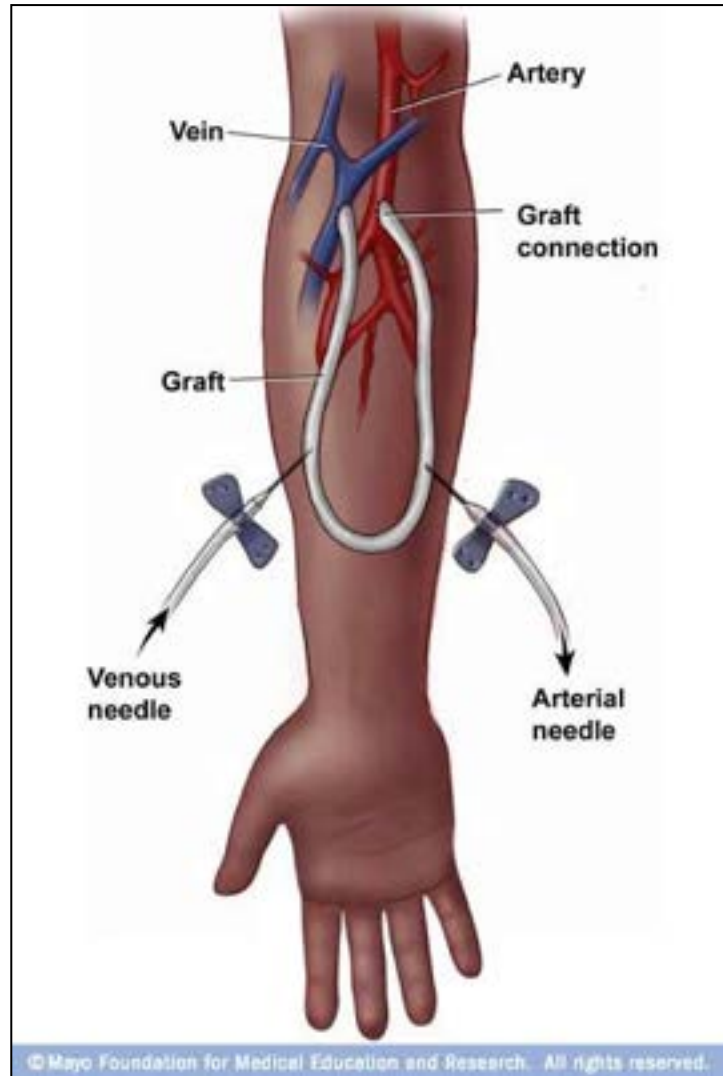


Fig. 87-1. **Standard radiocephalic AV fistula at the wrist.** Anastomosis of end of vein to side of artery.

AV fistula



AV Graft



AV graft

Tunneled cuffed catheter

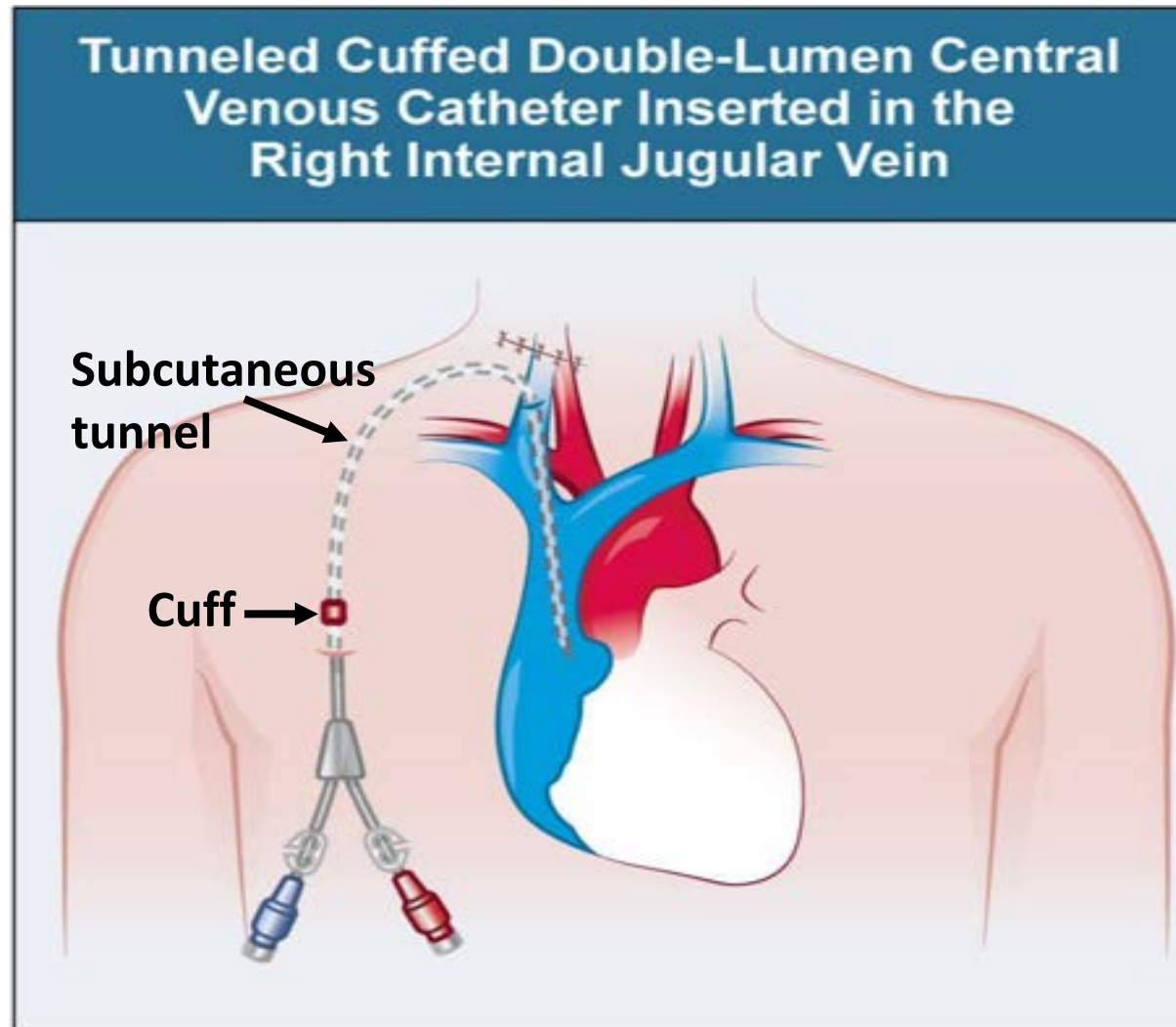


Fig. 87-11. Tunneled cuffed double-lumen central venous catheter inserted in the right internal jugular vein.

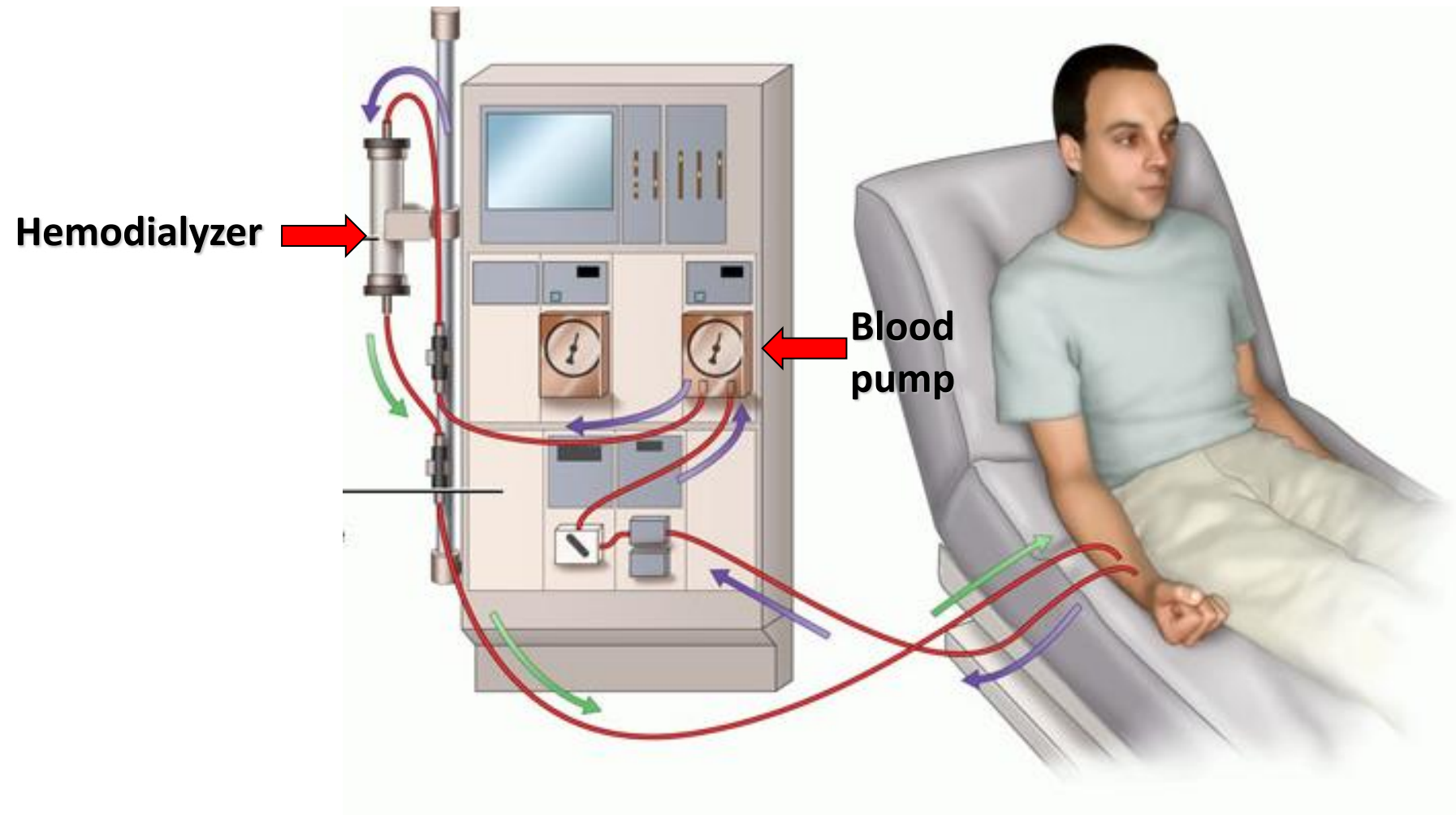
Tunneled cuffed catheter



Vascular access

- **Arteriovenous fistula (AVF) is the vascular access of choice for long-term HD because the risk of infection is the lowest**
- **Tunneled cuffed catheter has the highest risk of infection**

HD machine



Hemodialyzer

- Hemodialyzer in a HD machine is where the hemodialysis actually occurs



Two main types of HD

- In-center hemodialysis
- Home hemodialysis

In-center hemodialysis

- For in-center hemodialysis, the patients receive their hemodialysis treatment in a Hemodialysis Center, which is usually located inside a Hospital



Home hemodialysis

- A selected group of patients (usually younger and have day-time job) may be trained to perform their hemodialysis treatment by themselves at home



Comparison between In-center & Home HD

		In Center HD	Nocturnal home HD
	Location of HD	HD Center	Patient’s home
	HD performed by	Renal nurses	Patient
	Frequency of HD	2-3x/week	Depends on type of home HD machine used
	Duration of HD	4-5 hours / session	
➡	Dialysis efficacy	Higher in home HD	
➡	Quality of life	Better in home HD	
➡	Cost	\$\$	\$\$
➡	Need to train patient	No	Yes

Complications of HD

- 1. Complications related to vascular access**
- 2. Complications related to catheter access**
- 3. Complications during hemodialysis**

Vascular access complications

- Vascular access stenosis
- Vascular access thrombosis
- Upper limb ischemia

AV fistula stenosis

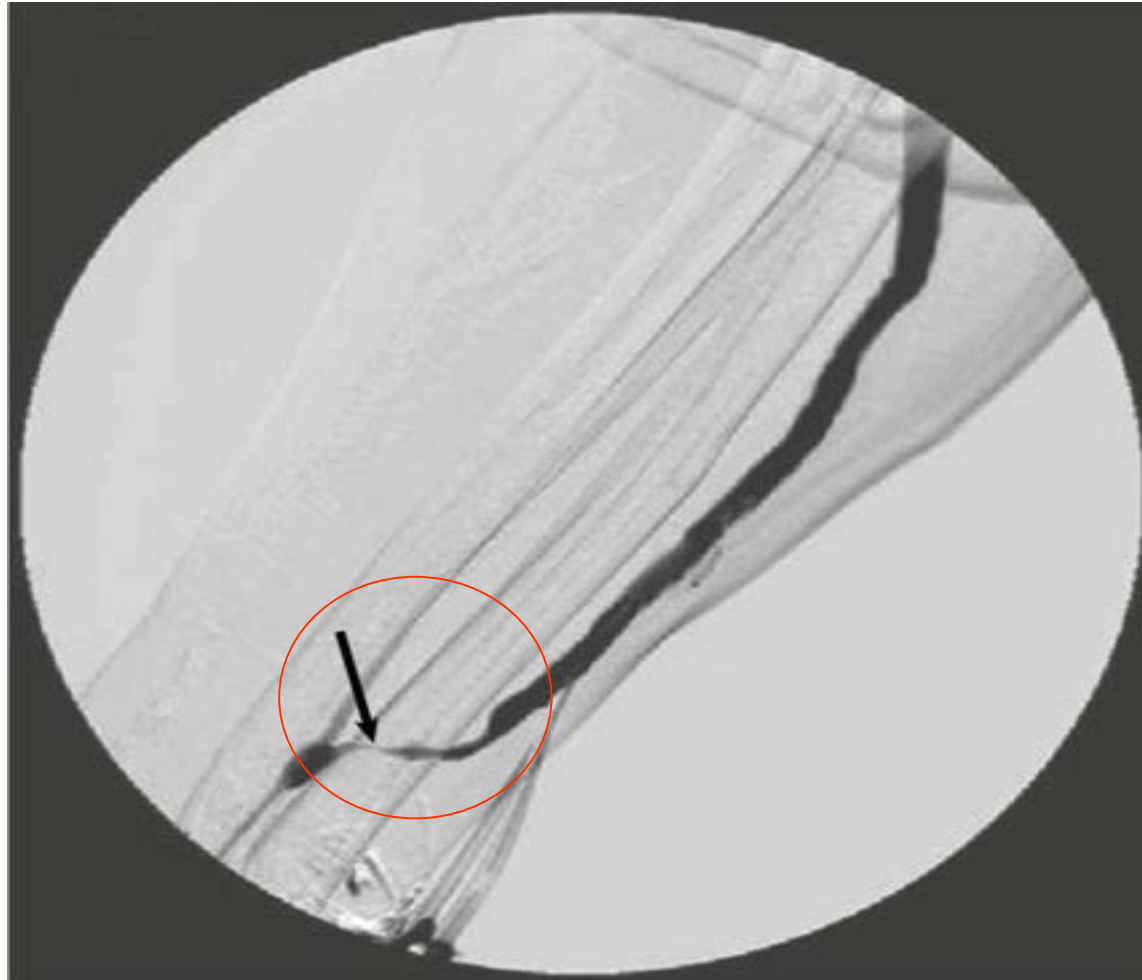


Fig. 88-9. Juxta-anastomotic stenosis in a radiocephalic arteriovenous fistula. Contrast material was injected at the arterial anastomosis (bottom left of image) and demonstrates a narrowing in the initial portion of the fistula (arrow).

Upper limb ischemia

- **Elderly and diabetic patients with severe atherosclerosis are prone to develop ischemia in the limb bearing the AV fistula.**



Catheter access complications

- Catheter exit site infection
- Catheter associated bacteremia
- Catheter malfunction

HD Catheter exit site infection

- Usually localized to the skin and soft tissue around the catheter exit site

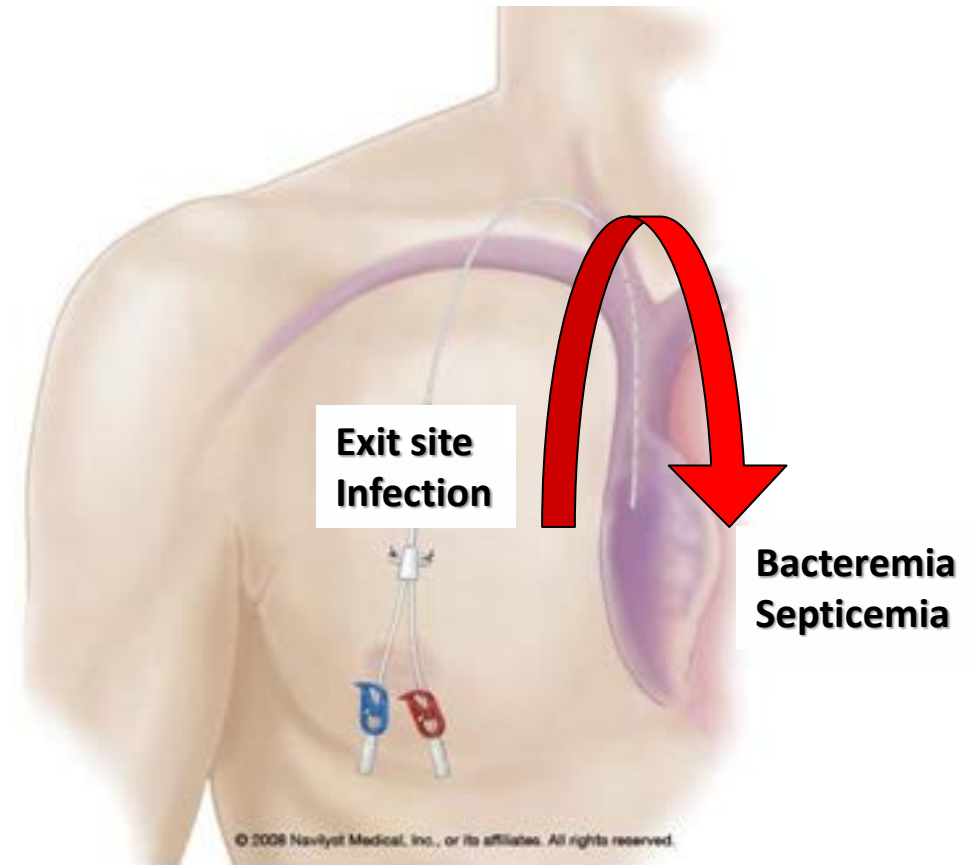


Courtesy Dr. T. M. Leitzig, University Hospital Erlangen, Germany

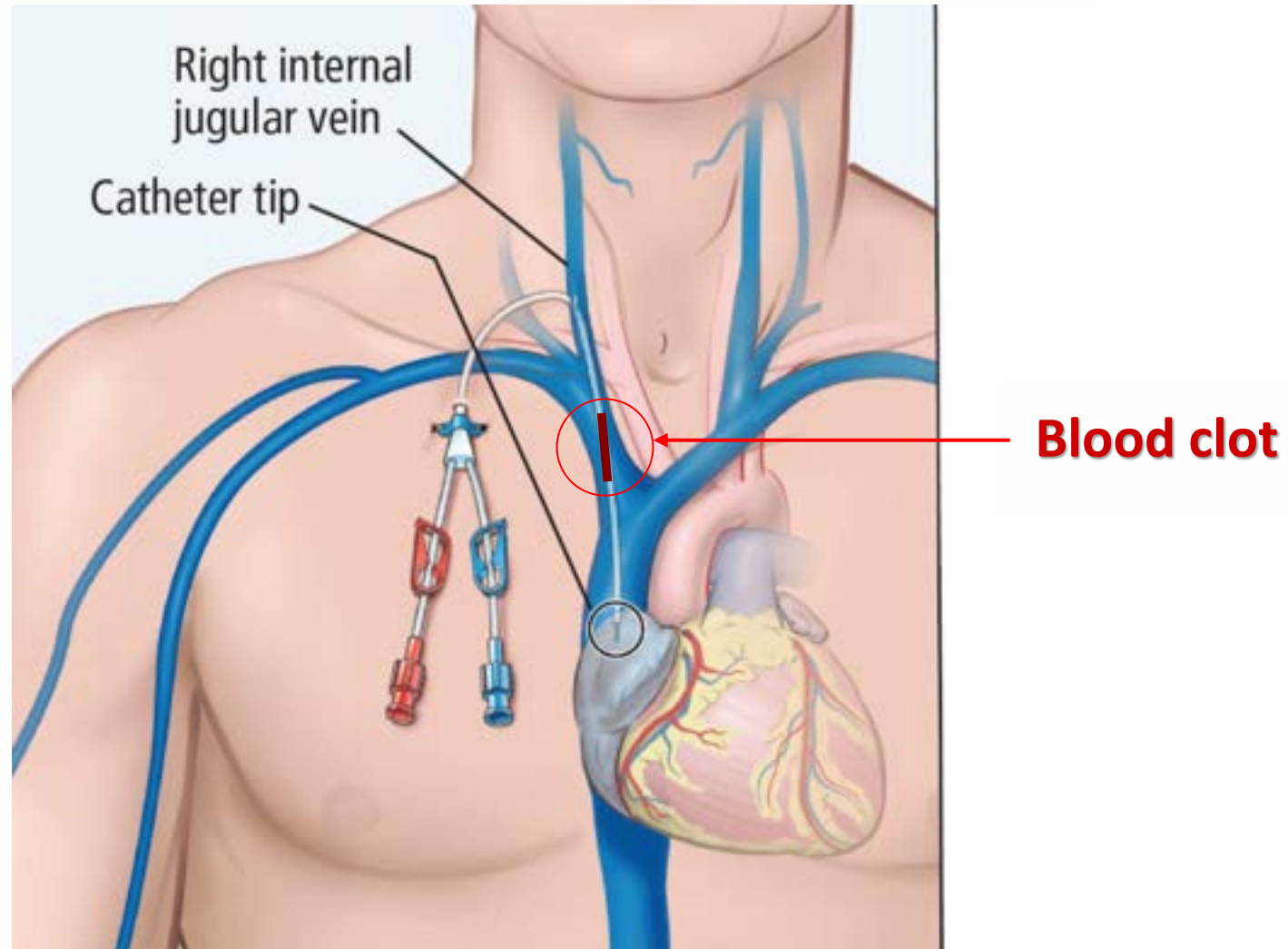
Fig. 87-13. Dialysis catheter tunnel infection.

HD Catheter exit site infection

- **Uncontrolled exit site infection may spread along the subcutaneous tunnel into the systemic circulation, causing bacteremia or septicemia**



HD Catheter malfunction

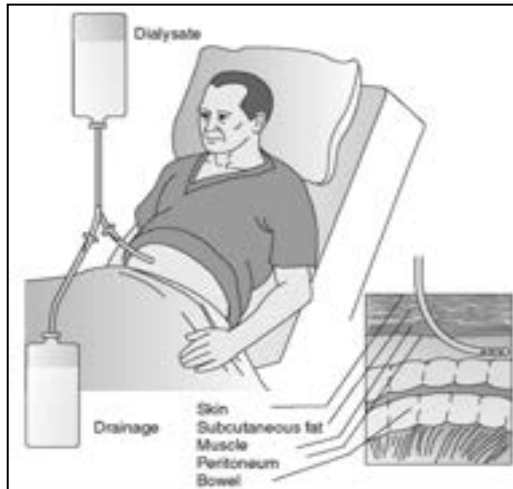


Complications during HD

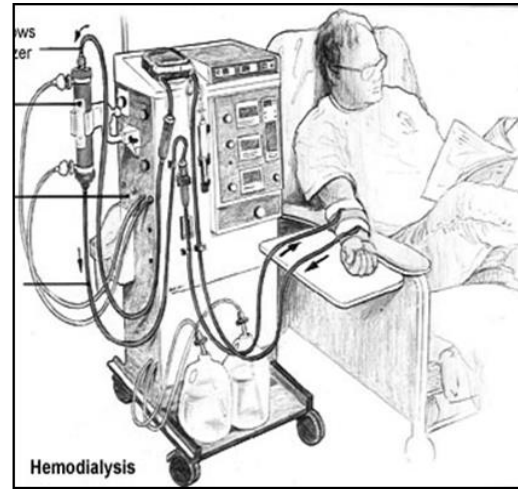
- **Cardiovascular**
 - Hypotension
 - Cardiac arrhythmia
- **Neuromuscular**
 - Muscle cramps
- **Technical malfunctions**
 - Air embolism
 - Blood loss



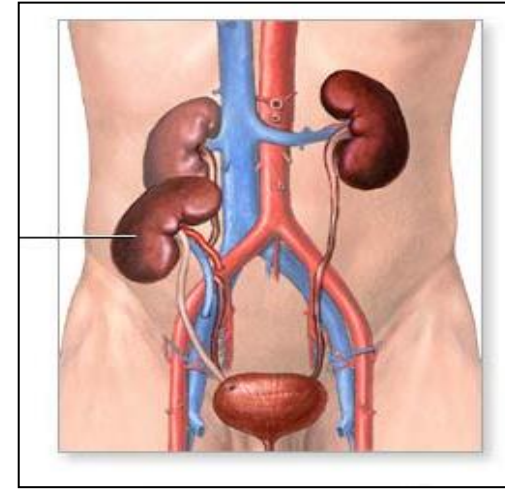
Three main forms of RRT



**Peritoneal Dialysis
(PD)**



**Hemodialysis
(HD)**



**Kidney
Transplantation**

Kidney Transplantation

- Kidney transplantation is the treatment of choice for most patients with ESKD



Two types of kidney donors

- There are 2 types of kidney donors, namely living donors and deceased donors



Living kidney donors

- In Hong Kong, the living donors are usually first-degree relatives or spouse of the patients
- The living donors should be healthy and have normal renal function & structure



Deceased kidney donors

- Deceased kidney donors have irreversible brain damage leading to brain stem death
- They should have no known history of kidney diseases, long-standing HT or DM and have normal renal function



Donor nephrectomy

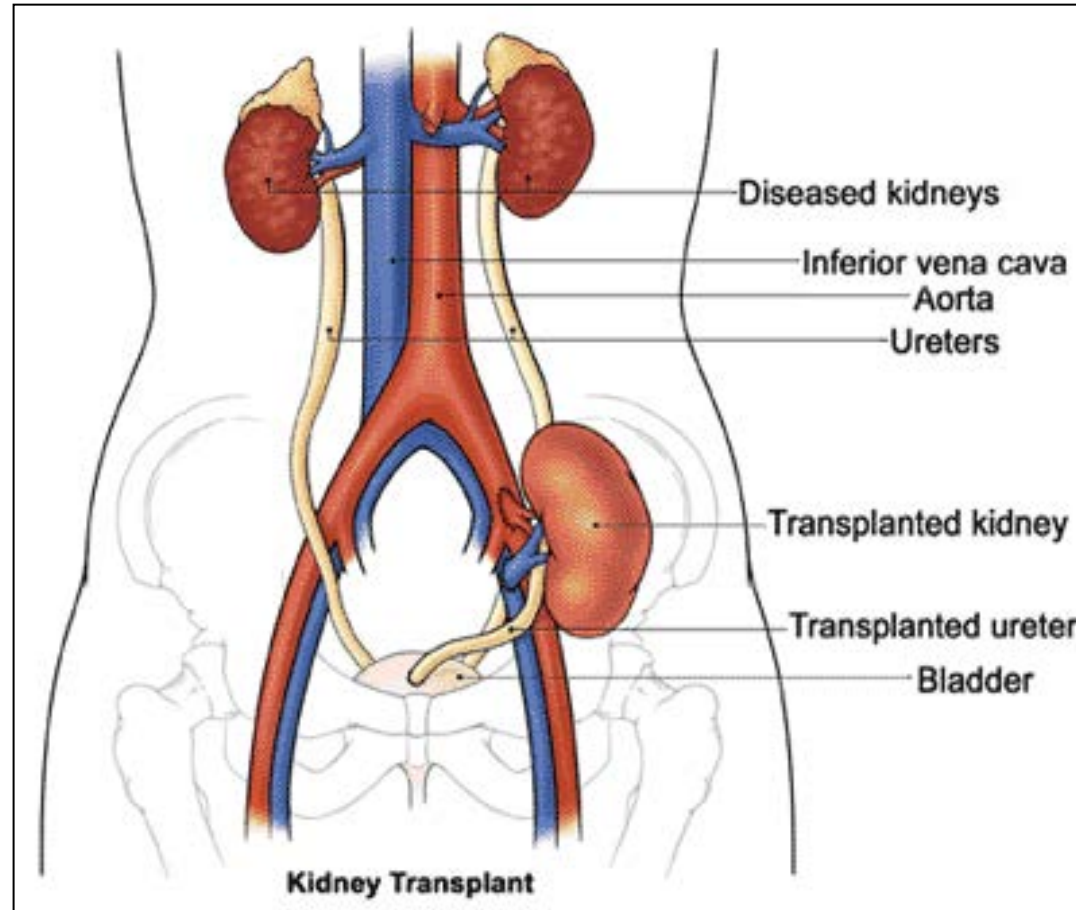


**Deceased donor:
Open nephrectomy**



**Living donor:
Laparoscopic nephrectomy**

Kidney transplant operation

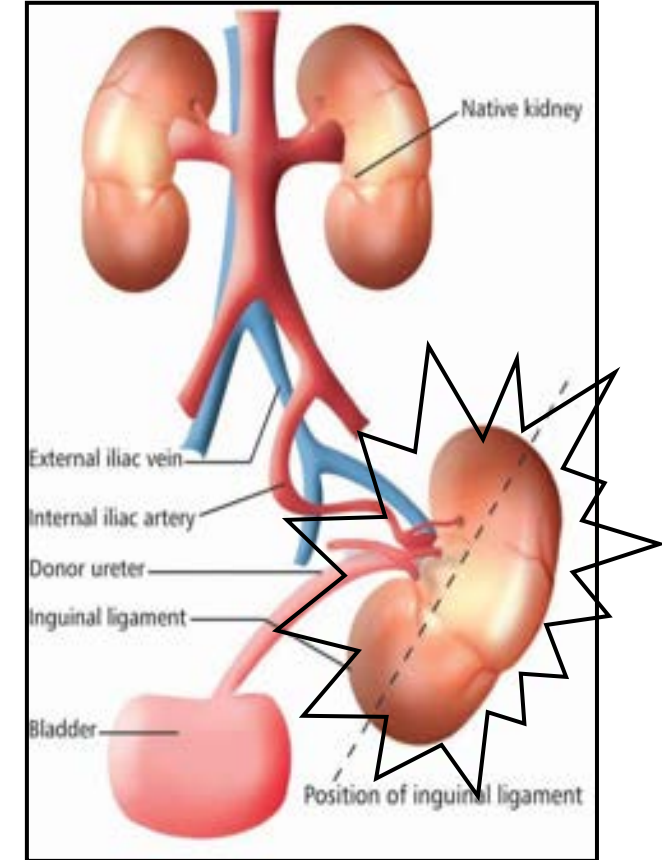


Kidney transplant scar



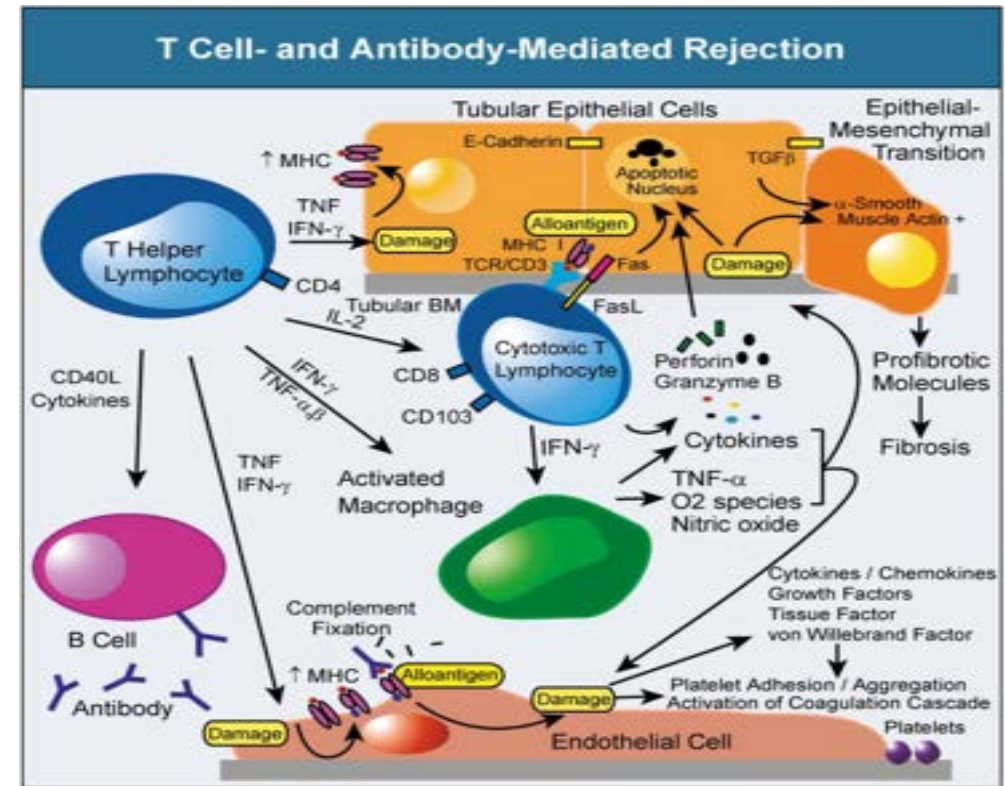
Acute rejection

- The main hurdle of successful kidney transplantation is acute rejection of the transplanted kidney by the host's immune system



Mechanisms of transplant rejection

- Acute rejection of the transplanted kidney involves both T cell and antibody mediated mechanisms



Rejection prophylaxis

- Long-term immunosuppressive treatment is needed to prevent rejection of the transplanted kidney by the recipient's immune system
- Risk of rejection is highest in the first 3 to 6 months after transplantation

Rejection prophylaxis

- **The degree of immunosuppression is therefore the heaviest during the first 3 to 6 months post transplantation**
- **Several immunosuppressive drugs are used concurrently in order to minimize the side effects of individual drug while achieving adequate level of immunosuppression**

Immunosuppressive drugs

- **Prednisolone**
- **Calcineurin inhibitors (CNIs): e.g. Cyclosporine, Tacrolimus**
- **Mycophenolate mofetil**
- **mTOR inhibitors: e.g. Sirolimus, Everolimus**
- **Azathioprine**

Side effects of immunosuppressive drugs

Drug	Common side effects
Prednisolone	Cushingoid facies, hyperglycemia, hyperlipidemia, peptic ulcers, hypertension
Cyclosporine	Nephrotoxicity, hypertension, hyperlipidemia, gum hyperplasia, gout
Tacrolimus	Nephrotoxicity, hypertension, new onset DM, tremors, GI upsets
Mycophenolate mofetil	Diarrhea, vomiting, anemia, leukopenia, viral infections
Sirolimus / Everolimus	Synergistic nephrotoxicity with CNIs, proteinuria, hyperlipidemia, impaired wound healing, interstitial pneumonitis
Azathioprine	GI upsets, hepatotoxicity, anemia, leukopenia, thrombocytopenia

Biologic agents

- **Biologic agents are polyclonal antibodies or monoclonal antibodies, used either as induction therapy or for the treatment of acute rejection**
- **Examples: anti-thymocyte globulin (ATG) or monoclonal anti-IL-2 receptor antibodies**

Causes of early allograft dysfunctions

- 1. Urological complications (e.g. urine leakage)**
- 2. Vascular complications (e.g. renal vein thrombosis)**
- 3. Infections (e.g. UTI, viral infection)**
- 4. Drug toxicity (e.g. calcineurin inhibitor nephrotoxicity)**

Long term complications

- 1. Infections**
- 2. Malignancy**
- 3. Cardiovascular disease**
- 4. Drug-related side effects**
- 5. Chronic allograft injury**
- 6. Recurrence of primary disease**

Infections

- **Renal transplant recipients are immuno-compromised & are prone to opportunistic infections**
- **Causative organisms include cytomegalovirus (CMV), pneumocystis jiroveci, BK virus and M. tuberculosis**

Pneumocystis pneumonia (PCP)

- **Usually present with acute onset of fever, cough and shortness of breath, associated with severe type I respiratory failure**
- **CXR shows diffuse, bilateral, interstitial infiltrates**

Pneumocystis pneumonia



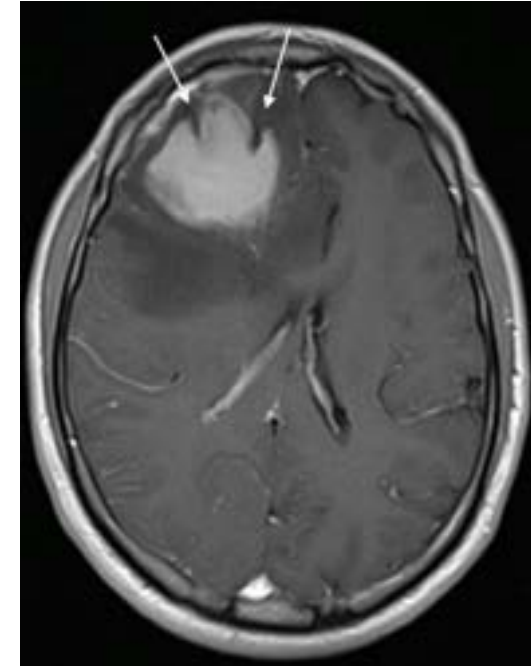
pneumocystis jiroveci pneumonia

Pneumocystis pneumonia

- **PCP is usually diagnosed by microscopy with staining of the broncho-alveolar lavage fluid for pneumocystis jiroveci**
- **PCP is treated with high dose septrin**
- **Transplant recipients are usually given septrin prophylaxis to prevent the development of PCP**

Malignancy

- Renal transplant recipients have increased risk of developing malignancies, including post transplant lymphoproliferative disorders (PTLDs)



**PTLD in
the brain**

Cardiovascular diseases

- **Cardiovascular disease is the most important cause of death with a functioning graft in renal transplant recipients**
- **Renal transplant recipients have multiple cardio-vascular risk factors such as diabetes, hypertension & hyperlipidemia (related to the underlying renal disease or the use of immunosuppressive drugs)**

Drug-related side effects



**Cushingoid features
(prednisolone)**



**Gum hypertrophy
(Cyclosporine A)**

Chronic allograft injury

- A clinico-pathological syndrome characterized by slow progressive decline in renal graft function, proteinuria, hypertension & histological features of interstitial fibrosis & tubular atrophy, which is caused by both immune & non-immune factors
- It is the leading cause of late graft failure after renal transplantation

Recurrence of primary disease

- **The cause of the renal transplant recipient's original kidney disease may recur in the transplanted kidney**
- **Recurrence of IgA nephropathy, membranous nephropathy & FSGS are not uncommon**

Summary-Renal replacement therapy

- **Three types of RRT, namely peritoneal dialysis, hemodialysis & renal transplantation**
- **Indications for initiation of RRT**
- **Choice between HD & PD**

Summary-Peritoneal dialysis

- **Essential components of PD**
- **CAPD vs. APD**
- **Complications of PD, especially PD related peritonitis**

Summary-Hemodialysis

- **Essential components of HD**
- **In center vs. home HD**
- **Complications of HD, especially those related to vascular & catheter accesses**

Summary-Transplantation

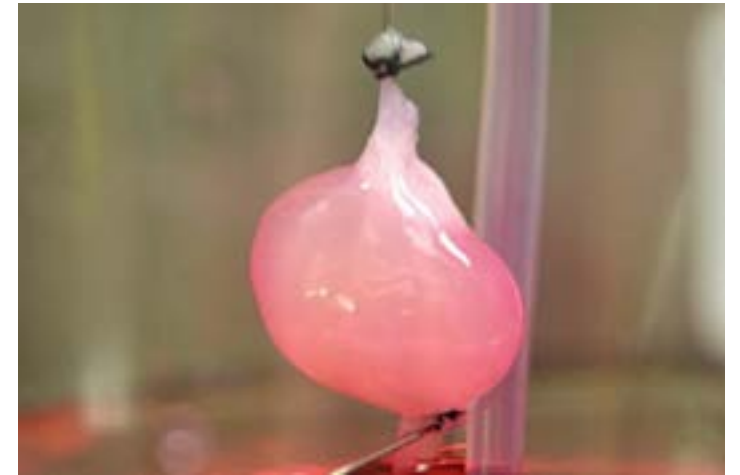
- **Basic principles of the transplant operation**
- **Acute rejection & principles of immunosuppression**
- **Early allograft dysfunctions**
- **Long term complications**

Looking ahead

- **Kidney transplantation remains the treatment of choice for ESRD patients**
- **Widespread use of kidney transplantation is limited by the shortage of donor kidneys**

Looking ahead

- **Advances in xeno-transplantation, gene editing technology and regenerative medicine might offer hope to solve the organ shortage problem**



Terminally ill man from Maryland, 58, is living with a PIG'S HEART - after becoming second patient ever to get pioneering transplant surgery

- A man in heart failure is the second-ever to receive a transplanted pig heart
- He is now breathing on his own, though his prognosis remains unclear
- **READ MORE:** First patient to receive [genetically-altered PIG heart transplant dies](#)

By [EMILY JOSHU HEALTH REPORTER FOR DAILYMAL.COM](#)

PUBLISHED: 21:37 GMT, 22 September 2023 | UPDATED: 23:16 GMT, 22 September 2023



Lawrence Faucette, 58, is the second person in the world to receive a heart transplant from a genetically modified pig. He was deemed ineligible for a human heart due to peripheral vascular disease, which reduces blood circulation. Pictured before surgery with his wife Ann, Mr Faucette is now breathing on his own, and his heart is functioning without any supportive devices

Research team reports longest successful transplant of a pig kidney into a human

By Deidre McPhillips, CNN

3 minute read · Published 11:01 AM EDT, Thu September 14, 2023



Dr. Jeffrey Stern, left, and Dr. Robert Montgomery, prepare to implant a genetically engineered pig kidney into the recipient's abdomen. Joe Carrozza for NYU Langone Health